ASR A2.2

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The Technical Rules for Workplaces (ASR) reflect the state of the art, occupational medicine and hygiene, as well as other established scientific findings for the safety and health of employees when setting up and operating workplaces.
They are determined or adapted by the
Committee for Workplaces
and published by the Federal Ministry of Labour and Social Affairs in the Joint Ministerial Gazette.
Within its scope, this ASR A2.2 specifies requirements of the Workplace Ordinance. If this Technical Rule is complied with, the employer can assume that the corresponding requirements of the Ordinance are fulfilled. If the employer chooses another solution, it must achieve at least the same safety and the same protection of employees' health.

Contents

1 Objective
2 Scope
3 Definitions
4 Suitability of fire extinguishers and extinguishing agents
5 Equipment for all workplaces
6 Equipment of workplaces with increased fire hazard
7 Organisation of operational fire protection
8 Deviating/additional requirements for construction sites
Annex 1 Standard scheme for determining the necessary fire extinguishing equipment
Annex 2 Examples for determining the basic equipment
Annex 3 Examples of deviations from the basic equipment
References

- Committee for Workplaces - ASTA Office - BAuA - www.baua.de -

1 Objective

This ASR specifies the requirements for equipping workplaces with fire alarm and fire extinguishing equipment, as well as the related organisational measures for operation, pursuant to section 3a paragraph 1, section 4 paragraph 3 and section 6 paragraph 3, including points 2.2 and 5.2 paragraph 1 g of the Annex to the Workplace Ordinance.

2 Scope

(1) This ASR applies to the setting up and operation of workplaces with fire extinguishing equipment, and to further measures for detecting, alarming and fighting incipient fires.
(2) For all workplaces pursuant to section 2 of the Workplace Ordinance, the requirements and design notes under point 5 of this rule (basic equipment) apply.
(3) For workplaces with normal fire hazard, the basic equipment is sufficient.
(4) For workplaces with increased fire hazard, additional measures under point 6 of this rule are required beyond the basic equipment.

Note:
For barrier-free design of measures against fires, ASR V3a.2 "Barrier-free design of workplaces", Annex A2.2: Additional requirements for ASR A2.2 "Measures against fires" applies.

3 Definitions

3.1 Fire hazard exists when combustible substances are present and there is the possibility of a fire starting.
3.2 Normal fire hazard exists when the probability of a fire starting, the speed of fire spread, the substances released and the associated danger to persons, the environment and property are comparable to the conditions of office use.
3.3 Increased fire hazard exists when

- flammable or oxidising substances or mixtures are present,
- the local and operational conditions are favourable for a fire to start,
- rapid fire spread or large smoke release is to be expected in the initial phase of a fire,
- work with a fire hazard is carried out (e.g. welding, flame cutting, cutting-off grinding, soldering) or processes are used that involve a fire hazard (e.g. paint spraying, flame work), or
- increased hazards exist, e.g. from self-heating substances or mixtures, substances of fire classes D and F, combustible dusts, extremely or highly flammable liquids or flammable gases.

Note:
Increased fire hazard within the meaning of this ASR includes the increased and high fire hazard according to the Technical Rule for Hazardous Substances TRGS 800 "Fire protection measures".
3.4 Incipient fires within the meaning of this rule are fires with so little smoke and heat development that persons can still approach safely with a clear view of the source of the fire.

3.5 Fire detectors serve for the early detection of fires and the triggering of an alarm. A distinction is made between automatic and non-automatic fire detectors (manual call points).
3.6 Fire extinguishing equipment within the meaning of this rule means portable or mobile fire extinguishers, wall hydrants and other hand-operated devices for fighting incipient fires.
3.7 Extinguishing capacity describes the ability of a fire extinguisher to extinguish a standardised test fire.
3.8 Extinguishing agent unit (LE) is an established auxiliary quantity that makes it possible to compare the performance of different types of fire extinguisher and, by addition, to determine the total extinguishing capacity of several fire extinguishers.
3.9 Fire safety helpers are the employees whom the employer has appointed for the task of fighting incipient fires.

4 Suitability of fire extinguishers and extinguishing agents

4.1 Fire classes

Fire extinguishers or extinguishing agents are assigned by the manufacturer to one or more fire classes according to their suitability. This assignment is shown on the fire extinguisher with pictograms (see Table 1).

Table 1: Fire classes according to DIN EN 2:2005-01 "Fire classes", pictograms according to DIN EN 3-7:2007-10 "Portable fire extinguishers - Part 7: Characteristics, performance requirements and tests"

Pictogram	Fire class
	Fire class A: Fires of solid substances (mainly of organic nature), normally burning with the formation of embers Examples: wood, paper, straw, textiles, coal, car tyres
	Fire class B: Fires of liquid or liquefiable substances Examples: petrol, oils, lubricating greases, paints, resins, waxes, tar Note: observe the safety data sheet
	Fire class C: Fires of gases Examples: methane, propane, hydrogen, acetylene, natural gas
	Fire class D: Fires of metals Examples: aluminium, magnesium, lithium, sodium, potassium and their alloys
	Fire class F: Fires of edible oils and fats (vegetable or animal oils and fats) in deep fat fryers and other kitchen equipment and appliances

For fires of electrical installations and equipment, DIN EN 2:2005-01 "Fire classes" does not specify a separate fire class.

Fire extinguishers according to DIN EN 3-7:2007-10 "Portable fire extinguishers - Part 7: Characteristics, performance requirements and tests" that are suitable for fighting fires in the area of electrical installations are marked with the maximum voltage and the necessary minimum distance, e.g. up to 1000 V, minimum distance 1 m.

4.2 Extinguishing capacity, extinguishing agent units

(1) The extinguishing capacity is indicated on the fire extinguisher by a combination of a number and a letter. In this combination, the number denotes the size of the standard test fire that was successfully extinguished and the letter denotes the fire class (see Figure 1).

[Figure 1 not reproduced]

Fig. 1: Example of the labelling of a fire extinguisher by the manufacturer, based on DIN EN 3-7:2007-10 "Portable fire extinguishers - Part 7: Characteristics, performance requirements and tests"

Notes:
1. The letters A, B, F denote the fire class for which the fire extinguisher is suitable. The numbers in front of them, 21A, 113B, 75F in Figure 1, indicate the extinguishing capacity in the respective fire class, determined on a standard test fire of corresponding size.
2. For fire classes A and B, it can be converted into extinguishing agent units (LE) with the help of Table 2.
3. For fire classes C and D, only the suitability of the fire extinguisher is determined, without determining the extinguishing capacity.
4. For fire class F, the number 75 in Figure 1 indicates that under test conditions a fire with a volume of 75 litres of edible fat/oil can be successfully extinguished. Fire extinguishers of fire class F are available with an extinguishing capacity of 5F, 25F, 40F and 75F. No conversion into extinguishing agent units (LE) takes place.
(2) Since the extinguishing capacity cannot be added up, an auxiliary quantity, the "extinguishing agent unit (LE)", is used to calculate the number of fire extinguishers required for fire classes A and B. A certain number of extinguishing agent units is thereby assigned to the extinguishing capacity of the fire extinguishers determined in the test, see Table 2. These values can then be added up per fire class.

Table 2: Assignment of extinguishing capacity to extinguishing agent units (assignment of fire extinguishers of the basic equipment according to point 5.2)

	Extinguishing capacity (rating according to DIN EN 3-7:2007-10)	
LE	Fire class A	Fire class B
1	5A	21B
2	8A	34B
3		55B
4	13A	70B
5		89B
6	21A	113B
9	27A	144B
10	34A	
12	43A	183B
15	55A	233B

(3) If fire extinguishers are provided for different fire classes, the extinguishing capacity must be sufficient for each of the fire classes present.

5 Equipment for all workplaces

5.1 Fire detection and alarm

(1) The employer must ensure by suitable measures that employees can be warned immediately in the event of a fire and called upon to leave buildings or endangered areas. The possibility of alerting emergency and rescue services must be ensured.
(2) Fires can be detected and reported by persons or by fire detectors. Fire detectors serve the early detection of fires. This contributes significantly to

the success of extinguishing and to the timely initiation of evacuation and rescue measures. Fire detectors are technical devices for triggering an alarm in the event of a fire. A distinction is made between automatic fire detectors, which detect a fire from its characteristics (e.g. smoke, temperature, flame), and non-automatic fire detectors, which are operated by hand (manual call points). The alarm can serve to warn the persons present or to summon help (e.g. security personnel, fire brigade).

(3) Suitable measures for alerting persons are, for example:

- fire alarm systems with voice alarm systems (SAA) or acoustic signal devices (e.g. horns, sirens),
- in-house alarm systems,
- electro-acoustic emergency warning systems (ENS),
- visual alarm devices,
- telephone systems,
- megaphones,
- hand sirens,
- calling out by persons, or
- personal warning devices.

(4) Technical measures must be implemented as a priority. Automatic fire detection and alarm equipment is to be preferred.

The need for technical alarm systems results from the risk assessment, e.g. if voice and visual contact or spatial conditions do not allow the endangered persons to be warned, or if a need for action arises from the evacuation drills according to ASR A2.3 "Escape routes and emergency exits" or from requirements of authorities.

5.2 Basic equipment with fire extinguishing equipment

(1) The employer must provide fire extinguishing equipment in sufficient number according to the type and quantity of combustible substances present in the establishment, the fire hazard and the floor area of the workplace. To determine the type and number of fire extinguishers required, the workplace can be divided into sub-areas if this is sensible or necessary because of the structural conditions or the conditions of use. The sub-areas belonging to a workplace can be classified into different fire hazards.

As a rule, the employer must provide fire extinguishers according to DIN EN 3-7:2007-10 "Portable fire extinguishers - Part 7: Characteristics, performance requirements and tests" as fire extinguishing equipment for the basic equipment. A general solution scheme for determining the equipment of the workplace is given in Annex 1; implementation examples for the basic equipment are shown in Annex 2 and for deviations from the basic equipment in Annex 3.

(2) In all workplaces, the number of fire extinguishers with the corresponding extinguishing capacity for fire classes A and B required for an area is to be determined for the basic equipment according to Tables 2 and 3. Based on the floor area (sum of the floor areas of all levels) of the workplace according to Table 3, the required extinguishing agent units are determined. The corresponding type, number and size of fire extinguishers according to their extinguishing capacity is then taken from Table 2, whereby the sum of the extinguishing agent units must at least correspond to the figure taken from Table 3 for each fire class.

Outdoor areas (e.g. green spaces, traffic routes) can be disregarded when determining the basic equipment.

Table 3: Extinguishing agent units as a function of the floor area of the workplace

Floor area up to ... m2	Extinguishing agent units (LE)
50	6
100	9
200	12
300	15
400	18
500	21
600	24
700	27
800	30
900	33
1000	36
per further 250	+6

As a rule, only fire extinguishers that each have at least 6 extinguishing agent units (LE) are counted towards the basic equipment.
By way of derogation, for normal fire hazard, fire extinguishers that each have only at least 2 extinguishing agent units (LE) can also be counted towards the basic equipment if:
- this simplifies operation, e.g. through a weight saving of at least 25 % per fire extinguisher,
- the access time is reduced, e.g. by halving the maximum distance to the nearest fire extinguisher according to point 5.3, and
- the number of fire safety helpers according to point 7.3 is doubled.

In multi-storey buildings, at least 6 extinguishing agent units (LE) must be provided on each floor.
To make portable fire extinguishers easy to handle, attention should be paid to
- a low device weight and
- within an area, the same mode of operation of the devices with regard to triggering and interruption devices.

Notes:
1. When selecting fire extinguishers, possible consequential damage caused by the extinguishing agents should also be considered.
2. When using carbon dioxide (CO_2) as an extinguishing agent, health hazards due to excessive CO_2 concentrations must be taken into account.

(3) If workplaces of different employers are located in a building, existing fire extinguishers can be shared. Each employer must ensure that its employees have access to the required fire extinguishers at all times.

5.3 Requirements for the provision of fire extinguishing equipment
The employer must ensure that in workplaces:
- fire extinguishers are mounted clearly visible and easy to reach,
- fire extinguishers are preferably mounted in escape routes, in the area of exits to the open air, at the access points to stairwells or at intersections of traffic routes/corridors,
- the distance from any point to the nearest fire extinguisher is no more than 20 m (actual walking distance), to ensure quick access,
- fire extinguishers are installed protected from damage and weather influences, e.g. by protective covers, cabinets, impact protection; this may be necessary, for example, at filling stations, in underground car parks or in structures not enclosed on all sides,
- fire extinguishers are mounted so that they can be removed from the bracket without difficulty; a grip height of 0.80 m to 1.20 m has proven appropriate,
- the locations of fire extinguishers are marked with the fire protection sign F001 "Fire extinguisher" according to ASR A1.3 "Safety and health protection signage". In workplaces that are not easy to oversee, the nearest location of a fire extinguisher is to be indicated clearly visibly by the fire protection sign F001 "Fire extinguisher" together with an additional "directional arrow" sign. Especially in long rooms or corridors, fire protection signs should be recognisable at all times in the direction of travel, e.g. by using flag or angle signs,
- other fire extinguishing equipment is likewise marked according to ASR A1.3 "Safety and health protection signage" (e.g. for wall hydrants: fire protection sign F002 "Fire hose"),
- the recognisability of the necessary fire protection signs on escape routes without safety lighting is maintained by using long-afterglow materials according to ASR A1.3, and
- the locations of the fire extinguishing equipment are included in the escape and rescue plan according to ASR A2.3 "Escape routes and emergency exits".

6 Equipment of workplaces with increased fire hazard
6.1 Determination of increased fire hazard
If areas with increased fire hazard are identified in the risk assessment, the employer must take additional establishment-specific and activity-specific measures (see point 6.2) in addition to the basic equipment according to point 5.2 and the basic requirements for provision according to point 5.3.
An increased fire hazard can be assumed, for example, in the following workplaces or for the following activities (see Table 4):

Table 4: Exemplary list of areas and activities in workplaces with increased fire hazard

1. Sales, trade, storage
   - Stores with extremely or highly flammable or easily ignitable substances or mixtures
   - Stores for recycling material and secondary fuels
   - Freight forwarding warehouses
   - Stores with paints and solvents
   - Waste paper stores
   - Cotton stores, timber stores, foam stores
   - Storage areas for packaging material
   - Stores with other combustible material
   - Furniture exhibitions
   - Sales rooms with increased fire hazards, e.g. DIY stores, hardware stores
2. Services
   - Cinemas, discotheques
   - Waste collection rooms
   - Kitchens
   - Accommodation establishments
   - Theatre stages
   - Technical and natural science areas in educational and research institutions
   - Tank and tanker cleaning
   - Dry cleaning, laundries
   - Old people's and nursing homes
   - Workshops for people with disabilities
   - Hospitals
3. Industry
   - Furniture manufacture, chipboard manufacture
   - Weaving mills, spinning mills
   - Paper manufacture in the dry section
   - Paper processing
   - Grain mills and animal feed production
   - Foam and roofing felt manufacture
   - Processing of combustible paints and adhesives
   - Painting and powder coating plants and equipment
   - Oil hardening shops
   - Printing works
   - Petrochemical plants
   - Processing of combustible chemicals
   - Leather and plastics processing
   - Plastic injection moulding
   - Cardboard box manufacture
   - Bakery products factory
   - Manufacture of machines and equipment
4. Trades and crafts
   - Motor vehicle workshop
   - Carpentry/joinery
   - Upholstery
   - Metalworking
   - Electroplating
   - Vulcanisation
   - Leather, imitation leather and textile processing
   - Bakery
   - Electrical workshop

6.2 Additional measures for increased fire hazard
Additional measures beyond the basic equipment in areas with increased fire hazard are, for example:
- equipping areas with fire alarm systems for the early detection of incipient fires,
- increasing the number of fire extinguishers and distributing them evenly in areas with increased fire hazard, in order to shorten the maximum distance to the nearest fire extinguisher and thereby the time until the start of fighting the incipient fire,
- mounting several fire extinguishers of the same type and construction at one location in areas with increased fire hazard, in order to achieve a greater extinguishing effect, if enough employees are present, by using several fire extinguishers at the same time to fight an incipient fire,
- providing additional fire extinguishing equipment suitable for the fire classes present on site in areas or at workplaces with increased fire hazard, in order to enable quick and effective fighting of incipient fires, e.g. carbon dioxide extinguishers in laboratories, fat fire extinguishers at fryers and deep fat fryers, mobile fire extinguishers with a greater throwing range and extinguishing performance at tank stores with flammable liquids, wall hydrants in buildings where a high extinguishing performance is needed for fighting incipient fires or for cooling, or
- measures that are necessary for activities with hazardous substances according to the Technical Rule for Hazardous Substances TRGS 800 "Fire protection measures".

(2) The extinguishing equipment to be used because of the increased fire hazard is to be arranged so that it can also be put into use quickly. Fire extinguishing equipment is therefore to be positioned in particular near the following places:
- processing machines with increased ignition hazard,
- increased fire loads, or
- rooms that are separated by fire protection measures because of the increased fire hazard.

It must be ensured that:
- the extinguishing agent is matched to the fire class,
- the amount of extinguishing agent is sufficient to cover an incipient fire of this hazard, and
- the fire extinguishing equipment is positioned so that, in the event of a fire breaking out in areas with increased fire hazard, it can still be reached quickly and without danger by the employee (as a rule not more than 5 m, at most 10 m actual walking distance).

(3) Fixed firefighting systems (e.g. sprinkler systems, water spray systems, fine water spray systems, foam, powder or gas extinguishing systems) are additional fire protection measures, i.e. measures beyond the basic equipment. They are required as a priority, for example, if:
- fighting the fire with fire extinguishing equipment is not possible because of the danger to oneself, or
- the areas are not accessible.

Note:
For activities with hazardous substances, the fire protection measures according to the Technical Rule for Hazardous Substances TRGS 800 "Fire protection measures" must be observed, and for the use of work equipment, the fire and explosion protection measures according to the Ordinance on Industrial Safety and Health.

7 Organisation of operational fire protection

7.1 Organisational fire protection measures

(1) The employer must determine and document the necessary measures against incipient fires, including the rules of conduct in the event of fire (e.g. evacuation of buildings).
Notes:
1. Information on the evacuation of buildings is contained in ASR A2.3 "Escape routes and emergency exits".
2. If the employer has identified an increased fire hazard, appointing a fire protection officer can be expedient. The officer advises and supports the employer on topics of operational fire protection. The need to appoint a fire protection officer can also result from other regulations, in particular state-law regulations.
(2) The measures for all persons present in the workplace are to be posted in a suitable form at an easily accessible place if:
- there is an increased fire hazard,
- posting an escape and rescue plan according to ASR A2.3 "Escape routes and emergency exits" is required, or
- visitors or external companies are frequently present in the workplace, especially if they are not accompanied.

This can be done, for example, as
- Fire safety regulations Part A according to DIN 14096:2014-05 "Fire safety regulations - Rules for drawing up and posting", or
- the "Rules of conduct in the event of fire" of the escape and rescue plan according to ASR A1.3 "Safety and health protection signage".

(3) The measures for all employees are to be made accessible to them by distribution or in electronic form. This can be done, for example, in the form of the Fire safety regulations Part B according to DIN 14096:2014-05 "Fire safety regulations - Rules for drawing up and posting".
(4) The measures for employees with special fire protection tasks, where these exist (e.g. fire protection officers), are to be made known to them against proof, where appropriate also electronically. This can be done, for example, in the form of the Fire safety regulations Part C according to DIN 14096:2014-05 "Fire safety regulations - Rules for drawing up and posting".

7.2 Instruction

The employer must instruct all employees on the measures determined under point 7.1
- before taking up employment,
- when the area of activity changes, and
- thereafter at appropriate intervals, but at least once a year.

7.3 Fire safety helpers
(1) The employer must familiarise a sufficient number of employees, through instruction and practice, with the use of fire extinguishing equipment for fighting incipient fires.
(2) The number of fire safety helpers results from the risk assessment. A share of 5 % of the employees is generally sufficient. A larger number of fire safety helpers may be necessary, e.g. in areas with increased fire hazard, when many persons are present, with persons with limited mobility, and when the workplace is large in spatial extent.
(3) When determining the number of fire safety helpers, shift work and the absence of individual employees, e.g. training, holidays and illness, must also be taken into account.
(4) Fire safety helpers are to be instructed competently with regard to their tasks. In addition to the basics of preventive fire protection, the content of the instruction includes knowledge of the operational fire protection organisation, the function and mode of operation of fire extinguishing equipment, the dangers from fires and the behaviour in the event of fire.
(5) Practical exercises (extinguishing exercises) in handling fire extinguishing equipment are part of the competent instruction of fire safety helpers.
Note:
In practice, with normal fire hazard, it has proven worthwhile to repeat the instruction with practice at intervals of 2 to 5 years. The time interval for the repetition is to be determined by the employer on the basis of the risk assessment.

7.4 Maintenance and testing

7.4.1 Fire alarm and fire extinguishing equipment
(1) The employer must have fire alarm and fire extinguishing equipment properly maintained at regular intervals, taking the manufacturer's information into account, and have it tested for functionality. The results are to be documented.
(2) If no defects are found, this is to be indicated on the fire extinguishing equipment, e.g. by attaching a maintenance record.
(3) If defects are found that mean the functionality of the fire extinguishing equipment is no longer ensured, the employer must arrange without delay for the fire extinguishing equipment to be repaired or replaced.

7.4.2 Special rules for fire extinguishers
(1) The components of fire extinguishers and the extinguishing agents contained in them can become unusable over time under the external influences at the place of installation (such as temperature, humidity, contamination, vibration or improper handling). To ensure functionality, fire extinguishers must therefore be serviced by a competent person every two years. If the manufacturer permits longer maintenance intervals than the period mentioned, the employer can use these. Shorter intervals specified by the manufacturer must be observed.
Notes:
1. Competent persons for the maintenance of fire extinguishers are in particular experts according to DIN 14406-4:2009-09 "Portable fire extinguishers - Part 4: Maintenance".

2. The periodic tests of the fire extinguishers (pressure test) by a competent person according to the Ordinance on Industrial Safety and Health remain unaffected by the maintenance carried out by the competent person.
(2) Under heavy use, e.g. due to environmental influences or mobile use, shorter intervals may be necessary.
Note:
For the necessary work steps, reference is made to the bvfa leaflet "Work steps for the maintenance of portable fire extinguishers", edition 2016-09 (01).

8 Deviating/additional requirements for construction sites
(1) The requirements in points 5.2 and 7.3 apply on construction sites only to stationary site facilities, e.g. site offices, accommodation or workshops.
(2) If activities with an increased fire hazard according to point 6.1 are carried out on construction sites, then for activities with a fire hazard (e.g. welding, flame cutting, cutting-off grinding, soldering) or when using processes that involve a fire hazard (e.g. paint spraying, flame work), one fire extinguisher for the corresponding fire classes with at least 6 LE must be kept ready in the immediate vicinity for each of the work equipment used there that causes an increased fire hazard.
(3) By way of derogation from point 7.3 paragraphs 1 to 3, all persons who work with the aforementioned work equipment are to be instructed theoretically and practically in the use of fire extinguishers according to point 7.3 paragraphs 4 and 5.
(4) Construction sites with special hazards (e.g. underground construction sites, high-rise construction) require additional measures against fires according to point 6.2.

Annex 1
Standard scheme for determining the necessary fire extinguishing equipment:
Step 1 - Determine the fire classes present according to Table 1
Step 2 - Determine the fire hazard (see also Table 4)
Step 3 - Determine the extinguishing agent units (LE) as a function of the floor area for the basic equipment with fire extinguishing equipment necessary in all workplaces, according to Table 3
Step 4 - Determine the number of fire extinguishing devices necessary for the basic equipment according to the extinguishing agent units (LE) in Table 2
Step 5 - If necessary, determine additional measures, in particular according to point 6.2, for increased fire hazard

Annex 2
Examples for determining the basic equipment

Example 2.1:
Office operation
Fire classes: A and B
Floor area: 500 m2
Result of the risk assessment: normal fire hazard
-> Basic equipment with fire extinguishers according to Table 3:
Table 3 gives 21 LE for up to 500 m2.

Powder extinguishers with an extinguishing capacity of 21A 113B are chosen, which according to Table 2 corresponds to 6 LE for this type of fire extinguisher.
21 LE divided by 6 means 4 fire extinguishers of this type are therefore required.

Example 2.2:
Day-care centre with 4 groups
Fire class: A
Floor area: 538 m2
Result of the risk assessment: normal fire hazard
Fire safety helpers: all employees are trained
-> Basic equipment with fire extinguishers according to Table 3:
Table 3 gives 24 LE for up to 600 m2.
As basic equipment according to point 5.2, a total of 24 LE would be required here, so with at least 6 LE per fire extinguisher, 4 fire extinguishers would be required.
The aim is that lighter fire extinguishers are available in each group, in the office and in the warming kitchen.
For the day-care centre, a total of 6 water extinguishers with 3 litres of water and an extinguishing capacity of 13A each, which according to Table 2 corresponds to 4 LE for fire class A for this type of fire extinguisher, are provided and positioned in the 4 groups, in the office and in the warming kitchen.
By selecting and positioning the fire extinguishers mentioned, the criteria of weight saving and reduction of the distance to the nearest fire extinguisher are met.

Example 2.3:
Kitchen with 3 deep fat fryers of 25 litres capacity each
Fire classes: A, B and F
Floor area: 700 m2
Result of the risk assessment: increased fire hazard

-> Basic equipment with fire extinguishers according to Table 3:
Table 3 gives 27 LE for up to 700 m2.
Powder extinguishers with an extinguishing capacity of 43A 233B are chosen, which according to Table 2 corresponds to 12 LE for this type of fire extinguisher.
27 LE divided by 12 means 3 fire extinguishers of this type are therefore required for the basic equipment.

-> Additional measures:
In addition, fat fire extinguishers with an extinguishing capacity of 75F are provided for the areas with fire class F.

Example 2.4:
Upholstery workshop
Fire classes: A and B
Floor area: 390 m2
Result of the risk assessment: increased fire hazard

-> Basic equipment with fire extinguishers according to Table 3:
Table 3 gives 18 LE for up to 400 m2.
Foam extinguishers with an extinguishing capacity of 21A 113B are chosen, which according to Table 2 corresponds to 6 LE for this type of fire extinguisher.
18 LE divided by 6 means 3 fire extinguishers of this type are therefore required for the basic equipment.

-> Additional measures:
In addition, an automatic fire alarm system is installed because of the work area that is not easy to oversee, as well as an extinguishing system.

Example 2.5:
Freight forwarding warehouse
Fire class: A
Floor area: 600 m2
Result of the risk assessment: increased fire hazard

-> Basic equipment with fire extinguishers according to Table 3:
Table 3 gives 24 LE for up to 600 m2.
Water extinguishers with an extinguishing capacity of 21A are chosen, which according to Table 2 corresponds to 6 LE for this type of fire extinguisher.
24 LE divided by 6 means 4 fire extinguishers of this type are therefore required for the basic equipment.

-> Additional measures:
In addition, 6 further water extinguishers with an extinguishing capacity of 13A are provided and distributed in the freight forwarding warehouse, in order to shorten the routes to the nearest fire extinguisher for even faster access.

Annex 3
Examples of deviations from the basic equipment
Applying the measures given in ASR A2.2 for determining the basic equipment of workplaces according to point 5.2 represents the appropriate solution for ensuring fire protection in a workplace.
Deviating from this determination of the basic equipment, the employer can choose another solution if it achieves at least the same safety and the same health protection for employees. This applies to normal as well as increased fire hazard.
The examples of such deviations listed in this annex show the procedure, but they neither replace the risk assessment nor do they represent a "model solution" that can be adopted without examining the specific conditions. Since deviations are possible on the condition that equivalence with the solutions according to ASR A2.2 is ensured, the deviations from ASR A2.2 must be determined and assessed. The employer must provide the proof of equivalence in the individual case on the basis of the risk assessment.

Examples for normal fire hazard
Example 3.1:
Administration
Fire class: A
Floor area: 600 m2, single storey
Result of the risk assessment: normal fire hazard
Note: a wall hydrant is present
According to point 5.2, a total of 24 LE would be required here, so with at least 6 LE per fire extinguisher, 4 fire extinguishers would be required as basic equipment.
The aim is to continue operating the existing wall hydrant and to take it into account as part of the basic equipment. The risk assessment also shows that

- water is suitable as an extinguishing agent,
- the floor is large enough (>400 m2 floor area) for the use of a wall hydrant to make sense,
- it is a wall hydrant with a dimensionally stable hose that can also be used by one person,
- a sufficient number of employees are instructed in handling this wall hydrant,
- smoke-logging of escape routes (e.g. stairwells) is avoided, because the wall hydrant is located in the corridor and its hose does not have to be led through fire or smoke protection doors to the source of the fire, and
- at least 2/3 of the required extinguishing agent units are covered by fire extinguishers, since wall hydrants should not be the sole fire extinguishing equipment.

Owing to its extinguishing capacity, up to 27 LE could be assigned to a wall hydrant. On the basis of the risk assessment, the existing wall hydrant is counted with 8 LE. The remaining 16 LE are covered by 3 powder extinguishers with an extinguishing capacity of 21A 113B, which according to Table 2 corresponds to 6 LE for this type of fire extinguisher.

Example for increased fire hazard
Example 3.2:
Kitchen operation with 3 small deep fat fryers with a filling quantity of 10 litres of edible oil each
Fire classes: A and F
Floor area: 180 m2
Result of the risk assessment: increased fire hazard
According to point 5.2, a total of 12 LE would be required here, so with at least 6 LE per fire extinguisher, 2 fire extinguishers would be required as basic equipment. In addition, fat fire extinguishers would be necessary because of fire class F.
To avoid confusion and duplicate equipment, the equipping should be done with fire extinguishers that are suitable for fire classes A and F. However, the available fire extinguishers only have an extinguishing capacity of 13A and 40F each, which according to Table 2 corresponds to 4 LE for fire class A for this type.

The risk assessment shows that
- all kitchen staff are trained as fire safety helpers,
- the probability that 2 fire extinguishers can be used at the same time is very high,
- the existing number of fire extinguishers can be reached very quickly, and
- even if the edible oil in the fryer re-ignites, further fire extinguishers (extinguishing agent reserve) can be put into use quickly.

On the basis of this risk assessment, a total of 3 fat fire extinguishers, also suitable for fire class A, with an extinguishing capacity of 13A and 40F each, which according to Table 2 corresponds to 4 LE for fire class A for this type, are positioned in the kitchen near the fryers.

References
- Technical Rules for Hazardous Substances (TRGS) 800 "Fire protection measures"
- DGUV Information 205-003 Tasks, qualification, training and appointment of fire protection officers 11/2014
- DGUV Information 205-023 Fire safety helpers 02/2014

- Committee for Workplaces - ASTA Office - BAuA - www.baua.de -